

Learning Journal

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1 2026-06-25 *Catching up: What I have learned during my first month*

A short paragraph on what you worked on. Inline math goes between dollar signs, like $f(x) = x^2$. Displayed equations get their own line:

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x). \quad (1)$$

You can point back to Equation (1) from anywhere.

A physical quantity with units, typeset properly: the acceleration due to gravity is about 9.81 ms^{-2} .

Flag something you want to find again:

Key result 1. The derivative is the limit of the difference quotient, when that limit exists. “Differentiable” means this limit exists at the point in question.

A few things I’m still unsure about:

- one open question
- another thing to revisit

A bit of code I was working through:

```
1 import numpy as np
2
3 def difference_quotient(f, x, h=1e-6):
4     return (f(x + h) - f(x)) / h
```

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Start writing here.